COMPUTER NETWORKS · CHAPTERS 1–5

500-word essentials
Dordal, An Introduction to Computer Networks (2nd ed.)

1 · BASICS

4 layers: Application → Transport (TCP/UDP) → IP → Link; each adds a header.

Datagram: (destination, next hop) table, stateless. **Virtual circuit:** connection ID, per-connection state.

TTL stops routing loops: −1 per router, drop at 0.

Delay = transmit + propagate (≈200 m/µs) + store-and-forward + queue; only queue is engineerable.

Most loss is **congestion**. **MAC** = 48-bit, no location. **IP** = 32-bit prefix + host (CIDR).

IP is best-effort/connectionless; fragmentation supported. **Switch** = LAN address; **router** = IP address.

TCP: reliable, ordered, ports, sliding window, Reno sawtooth. **UDP:** no setup/retransmit; real-time.

Socket = (host, port); **80** = web, **25** = email. **DNS** maps names to IPs.

Throughput = effective rate; **goodput** = useful app data. **kbps** = 10^3 bits; **kB** = 2^10 bytes.

2 · ETHERNET

CSMA/CD: listen, send, detect collision, random backoff. IEEE: **802.3** Ethernet, **802.11** Wi-Fi.

Slot 512 b = 64 B = 51.2 µs; min frame 64 B; max data **1500 B**; gap 96 b.

Backoff: attempt N → k < 2^N slots; give up at **16**.

Switch learns (source, port), forwards known, floods unknown; isolates collision domains.

Hub repeats and propagates collisions; a switch does not. Max **1024** hosts.

MAC flooding attack → **port security**. **ALOHA** max ≈18%; Ethernet 64–80%.

3 · ADVANCED ETHERNET

Spanning tree: root = smallest ID; BPDUs to 01:80:c2:00:00:00; prune loops; **~30 s**.

VLAN: 802.1Q tag (12-bit ID); ≈256 hosts; a router connects VLANs.

SDN: controller programs flow tables (match + action).

TRILL/SPB: keep all links active, route shortest paths; recovery **<0.1 s**.

4 · WIRELESS

Radio **cannot detect collisions**; **hidden nodes** collide at a third station.

CSMA/CA + link-layer ACK; slot **20 µs**, DIFS **50 µs**, SIFS **10 µs**.

RTS/CTS fixes hidden nodes; rate scaling 54→6 Mbps; **MIMO** uses multiple antennas.

WPA2 = AES/CCMP; Personal (PSK) vs Enterprise (802.1X). Four-way handshake: ANonce+SNonce → **PTK**; **KRACK** replays msg 3. **WPA3** = SAE.

WiMAX/LTE schedule uplink; **ranging** sets timing/power; **RACH** on entry. **Satellite:** GEO 35,786 km, RTT ≈500 ms.

2.4 GHz uses channels **1/6/11**; beacon every **100 ms**; ESS = same SSID roaming.

5 · OTHER LANs

VPN: tunneling + virtual interface.

Token Ring: token; collision-free and fair (1/N).

VC: VCI unique per link, rewritten per switch; QoS but stateful.

ATM: 48 B data + 5 B header; **AAL5** 10% overhead (one CRC-32), **AAL3/4** >20%.

FORMULAS

Next hop = neighbour on shortest path; direct if local.

Learn source+port; flood unknown.

Spanning tree: root = min ID; prune non-tree links.

Longest prefix wins; /0 last.

VCI: (VCI in, port in) → (VCI out, port out).

ATM cells = ceil(P/48) for AAL5.

Token wait = (N/2) × token time.

Contention = slots × 20 µs.

Delay = (1+R)×tx + km/200.

NUMBERS

1500 B Ethernet · **576 B** min IP · **64 B** min frame · **48+5 B** ATM · **51.2 µs** slot · **1024** hosts · **N=16** · **48-bit** MAC · **32-bit** IPv4 · **200 m/µs** · **30 s** STP · **10^4–10^5** switch entries.